

TECHNICAL PROPOSAL

HELI DECK COMMUNICATION SYSTEM

DOCUMENT ID: HSL_HDCS REV00

Date: 30th October 2020

Prepared by:



Elcome Integrated Systems Pvt. Ltd.

Plot No. EL-205, MIDC Mahape,

TTC Industrial Area, Navi Mumbai - 400710

(<http://www.elcome.in>)

HISTORY SHEET

Rev No.	Date	Changes	Cause	Implemented
00	30 Oct 20	All	First Issue	SK Singh

Elcome Integrated Systems Pvt. Ltd.

Plot No. EL-205, MIDC Mahape,

TTC Industrial Area, Navi Mumbai - 400710

(<http://www.elcome.in>)

All rights reserved. No part of the document may be reproduced or transmitted in any form or by any means, electronic or mechanical, for any purpose, without the written permission of Elcome Integrated Systems Pvt. Ltd.

Company or product names mentioned in this document may be trademarks or registered trademarks of their respective companies.

Table of Contents

Table of Contents..... 3

Chapter I: General..... 5

1. Preamble5

2. Identification of this document5

3. Scope of this document.....5

4. Documents referred.....5

5. General5

Chapter II: System Overview 6

1. Introduction.....6

2. Requirement Overview6

3. TETRA System Overview6

3.1. Frequency Band..... 7

3.2. Operation Philosophy..... 7

4. Salient Aspects of TETRA7

Chapter III: Technical Details..... 8

1. Introduction.....8

2. Building Blocks.....8

3. Technical Specification8

4. Features of the system.....8

5. Our Approach9

6. Base Station10

6.1. UHF TXRX-1.0 Features 10

6.2. Specifications 11

7. Handheld TETRA Radio (Handset and Headset)11

7.1. Multi Charger 13

8. Fixed units13

9. Voice Logger13

Table of Figures

Figure 1: Typical TETRA System 6

Figure 2 General System Architecture 9

Figure 3 TETRA Base station 10

Figure 4 Handheld radio (Portable) 12

Figure 5: Typical Helmet Mounted Headset 12

Chapter I: General

1. Preamble

Heli-Deck Communication System (HDCS) is installed on warships and similar critical afloat platforms to facilitate reliable communication during helicopter operations at sea. Robustness, efficiency and reliability of HDCS onboard are vital for successful accomplishment of flying operations.

2. Identification of this document

This document is identified as HDCS Rev XX where XX is the current revision number. This technical proposal refers to several external documents that are identified subsequently in this document.

3. Scope of this document

This document explains the system constituents and provides technical outline for delivery of one set of HDCS equipment.

4. Documents referred

The scope of work has been worked out on the basis of Hindustan Shipyard Limited RFQ reference 3400000297 dated 7th Sep 2020.

5. General

This document provides an overview of our design approach for the TETRA radio system for HDCS. The pictures and details provided in this document should be considered as proof of concept and our capability to meet the requirements, actual look and feel of final product may vary while remaining compliant to the technical specifications of the RFP.

Chapter II: System Overview

1. Introduction

The term TETRA stands for Terrestrial Trunked Radio. It is aimed to provide a private mobile radio communications system. TETRA has become popular due to its bouquet of advanced features which include group communications on simultaneous multiple channels with enhanced security features.

2. Requirement Overview

Before we dwell into the details of TETRA system implementation, let us capture the basic requirements as per the NSQR.

The requirement is to realise wired as well as wireless communication using contemporary TETRA for communications to ensure safe and efficient conduct of flying operations onboard a ship at sea.

The system would be COTS in nature.

The system should have priority allocation, multi-channel operation and capable of integration with shipboard communication and intercom systems. It should facilitate voice as well as data communication.

3. TETRA System Overview

TETRA uses Time Division Multiple Access (TDMA) with four user channels on one radio carrier with 25 kHz spacing between carriers. Both point-to-point and point-to-multipoint transfer can be used. Digital data transmission is also included in the standard though at a low data rate.

TETRA provides for authentication of terminals towards infrastructure and vice versa. For protection against eavesdropping, air interface encryption and end-to-end encryption is available.

TETRA is a unique combination of group voice communications, mobile telephony and mobile data services specifically designed for authority use.

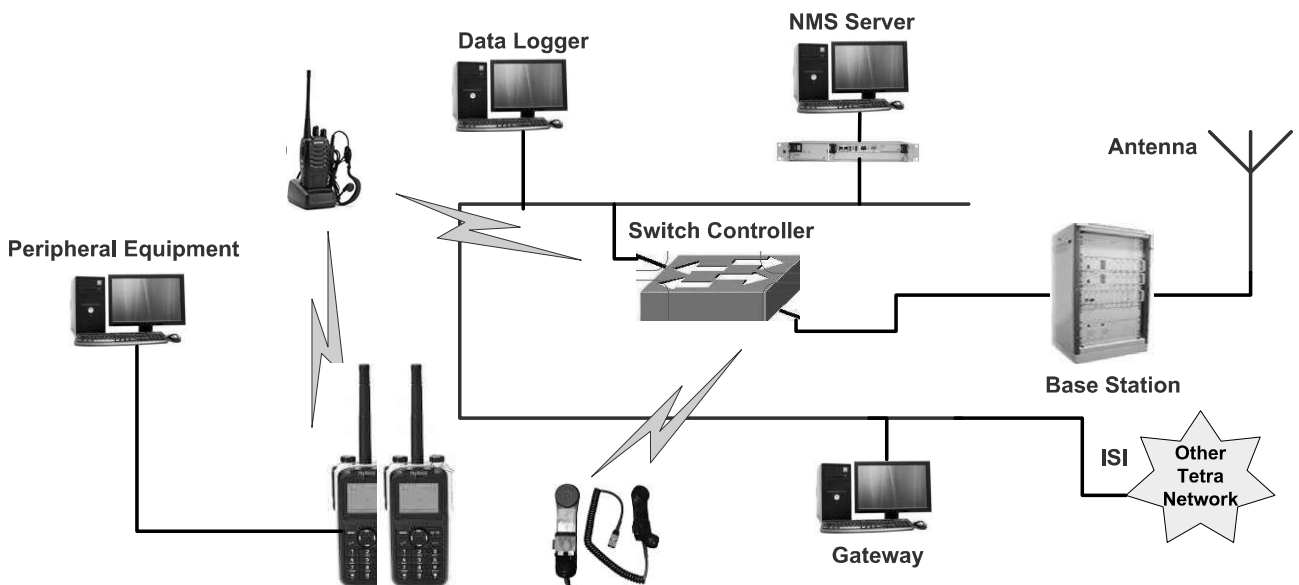


Figure 1: Typical TETRA System

3.1. **Frequency Band**

As per ETSI, the founders of TETRA, frequency band of 380 - 400 MHz is used for public safety and security and 410-430 MHz frequency has been reserved for the professional cellular sector.

3.2. **Operation Philosophy**

TETRA operates on the principle of Time Division Multiple Access (TDMA). It supports four RF channels having 25 KHz bandwidth and transmission rate of 7.2 kbps per slot. Each time slot occupies 14.157 ms. TDMA provides efficient RF communication with higher throughput even in adverse RF conditions, concurrent voice and data communication and full duplex voice communication.

4. **Salient Aspects of TETRA**

TETRA is gaining popularity as digital private communication network due to following characteristics:-

- a) High Quality and High Security in communications.
- b) Double spectral efficiency.
- c) Higher data bandwidths (up to 28.8 kbps)
- d) Spectral efficiency (4 time slots in 25 KHz)
- e) Multiple real-time data services (status, short data, circuit mode data, packet mode data).
- f) Half-duplex and full-duplex communications.
- g) Interoperable with other networks (TETRA, ISDN, IP, GSM).
- h) Continuous coverage.
- i) Emergency calls.
- j) Fast call setup (<300 ms)
- k) Simultaneous voice and data.
- l) Open standard technology: protects investment.
- m) Optimized for high population density areas
- n) Data bandwidth on demand (up to 28.8 kbps)
- o) Enhanced Security

Chapter III: Technical Details

1. Introduction

This chapter elaborates our approach for delivery of TETRA based Helo-Deck Communication System (HDCS).

2. Building Blocks

Based on the basic architecture explained in Figure 1, the building blocks of a TETRATETRA based communication system are as follows:-

- a) TETRA Hardware
 - i) Base Station
 - ii) Handset/Headset
 - iii) Antenna system
- b) Processing and Networking
 - i) Networking hardware
 - ii) Network Management System
 - iii) Data Logger
 - iv) Peripheral System Interface
 - v) Intersystem Interface

3. Technical Specification

The solution proposed by us will be based on following guiding features:-

Technology	:	TETRA Technology.
Frequency Range	:	V/UHF 400-430MHz.
Mode of communication:		Digital Encrypted voice
Frequency stability	:	≤1 ppm
Channels	:	3 Channels catering for 16 wireless positions.
RF Power Output	:	< 5W

4. Features of the system

The system will be designed to comply to the TETRA technology as mandated in the NSQR.

The system will be designed as modular system based on TETRA technology. This would facilitate adding of more nodes and base stations at any time.

The system consists of TETRA hardware comprising of Central processor, V/UHF Radio and Antennae on the deck. Additional unit may be considered as an option to provide full redundancy.

The system will operate on 230V, 1 Ph, 50/60HZ power supply of the ship. A UPS has been catered to provide uninterrupted power to all HDCS equipment. A back up of 20 minutes has been catered which is sufficient for emergency power to be made available or graceful shutdown to be carried out. All internal power supplies required for the system operation will be part of deliverables.

The headset and helmet mounted headset will be provided as specified. These will be provided with batteries and a battery charging unit will also form part of deliverables.

The system will permit various modes of operation. The basic mode will be the trunked mode of operation wherein all users will communicate using the base station and the UHF antenna mounted on upper deck. In Direct Mode Operation (DMO) two hand set users can communicate directly to each other.

All capabilities of TETRA ie status, short data, circuit mode data, packet mode data etc will be provided in the Voice Mode or Voice + Data mode as mandated.

5. **Our Approach**

We intend procuring the equipment from M/S Damm, Denmark and M/S Consort Digital who are manufacturers of repute for Tetra based systems. To their credit they have Tetra systems deployed in Delhi Metro, Mumbai Suburban Trains and many other places.

A simplified system architecture is presented below.

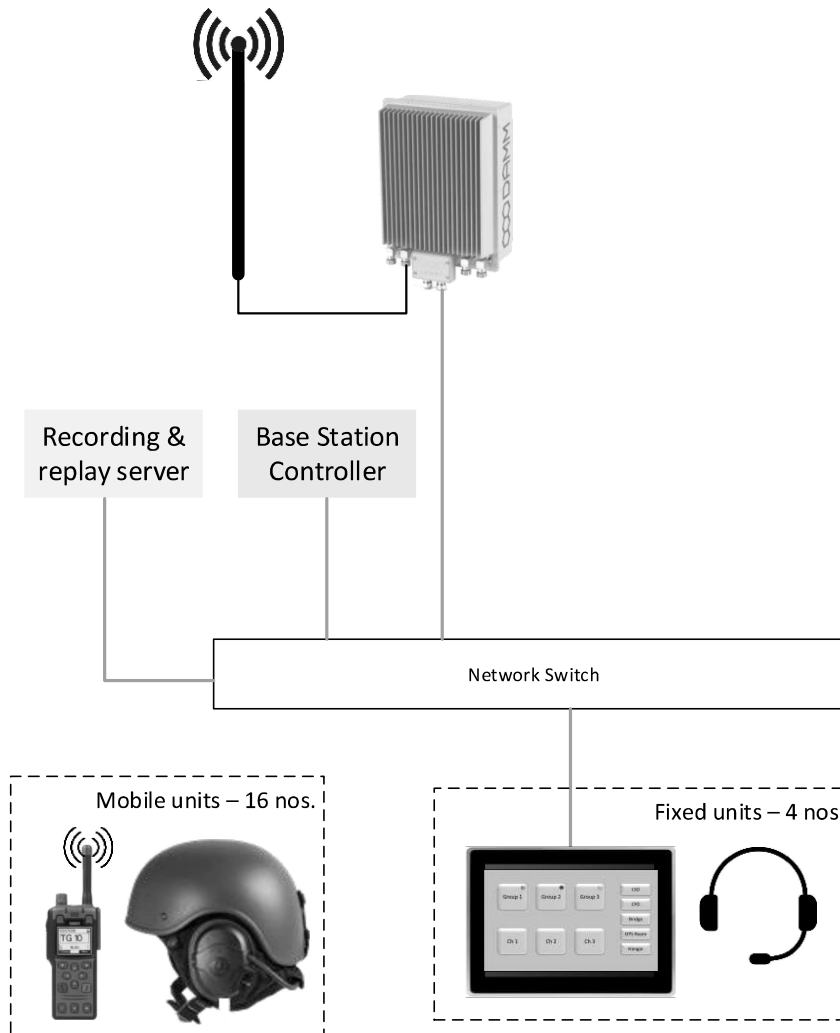


Figure 2 General System Architecture

6. **Base Station**

The base station is the heart of the TETRA system. It comprises of a UHF Transceiver and its control unit. The control unit will be realised using off the shelf server hardware. We propose to use DAMM BS421 or equivalent outdoor base station.

6.1. **UHF TXRX-1.0 Features**

The transceiver has the following features.

- **IP65-encapsulated TETRA base station**
The DAMM BS421 Outdoor System is a mast-mountable and compact IP65-encapsulated TETRA communication system. This advanced and powerful base station transceiver and distributed controller is ideal for outdoor mounting because of its compact size and its rugged design. Additionally, it is equally well suited for indoor use.
- **Reduce OPEX with low base station power consumption**
The compact design of DAMM's outdoor TETRA system enables direct mounting on antenna masts, buildings and towers, reducing installation costs considerably. Thanks to the low power consumption our base stations generate minimal heat, and together with the IP65 encapsulation this eliminates the need for cooling fans that require maintenance. Furthermore, redundancy means that maintenance can be done without shutting down operations.
- **Full network flexibility**
The DAMM TetraFlex system is scalable and can handle sites of any size – from one-site solutions with local coverage to large-scale multisite solutions, using any combination of outdoor and indoor DAMM base stations. The intelligent DAMM TetraFlex software enables simple and smooth site expansion, even while the system is in operation. The 100% IP-based technology provides full architectural network flexibility with all network components connected in a single state-of-the-art infrastructure without the need for a central switch.
- **Multi-technology integration**
This product is part of our single-technology portfolio, but can be fully integrated into any of our multi-technology products.
- **Easy to install – anywhere**
Plug-and-play simplicity enables easy setup and fast deployment. With its IP65 protection the DAMM Outdoor System is ideal for installation in harsh environments. A compact design enables direct mounting on antenna masts, buildings and towers, reducing feeder loss and installation costs considerably, while also providing an ideal solution for rapid, vehicle-based deployment.
- **Secure and failure tolerant**
Featuring full support for battery backup as well as redundant controllers, power supplies and carriers, DAMM TetraFlex meets your highest requirements on reliability. The intelligent distributed architecture replicates information to all sites in the network, avoiding any single point of failure.



Figure 3 TETRA Base station

- Unlimited application integration
DAMM TetraFlex is supplied with Application Gateways for easy access to the Application Programming Interface (API), Packet Data Gateways and Voice Gateways, allowing you straight-forward development of customer-designed applications or integration into existing telephone systems and control room equipment. As an independent TETRA infrastructure provider DAMM Base Stations are IOP certified for full compatibility with all TETRA terminals, providing you with the option to freely choose your favourite terminal brand.
- Wide range of advanced software tools
The DAMM TetraFlex system is available with a wide range of application software such as Voice and Data Log System, Dispatcher, Group Bridge and a comprehensive Network Management tool including subscriber management and network performance statistics.

6.2. **Specifications**

Transmitter and receiver

- TX power at antenna connector:
- 0.5W to 10W TETRA remotely adjustable
- RX sensitivity with diversity, static: -121dBm
- RX sensitivity without diversity, static: -118dBm
- RX sensitivity with diversity, dynamic: -118dBm
- RX sensitivity without diversity, dynamic: -112dBm
- Receiver diversity: Dual as standard
- Built-in duplexer:
- Combines TX and RX to one antenna
- Redundancy: Hot standby
- Synchronization: Internal or external with GPS

7. **Handheld TETRA Radio (Handset and Headset)**

Built to withstand the very harshest working conditions, the Intrinsically Safe STP8X100 or equivalent delivers clear audio and enhances safety for workers in hazardous environments.

The reduced-keypad is perfect for use with gloved hands and with an IP67 environmental protection rating - this intrinsically safe radio will continue to work after submersion in up to 1m of water for up to 30 minutes.

The STP8X100 meets the exacting requirements of the IECEx/ATEX standards.

In locations where normal TMO network coverage is unavailable or unreliable, users can switch their radios to Direct mode to maintain communication.

Key Features:

- Voice calls (Encrypted and Clear)
- Status and SDS message
- Emergency calls
- Call Participation and Call Monitoring

SECURE

Designed for mission critical use, the STP8X100 supports ETSI and SFPG recommended Air Interface Encryption and End to End Encryption

ATEX STANDARD

Meets the requirements of the IECEx/ATEX/Hazardous location standards

Storage Expansion

MicroSD card slot enables further storage

ADVANCED SAFETY FEATURES

Protection and reassurance for lone or remote workers. Man-Down offers motion- and tilt-monitoring, local alarm and over-the-air alerts for vulnerable users working in dangerous environments



Figure 4 Handheld radio (Portable)



Figure 5: Typical Helmet Mounted Headset

Figure 5 above depicts a typical helmet mounted headset. It will connect to the handset using the cable.

7.1. **Multi Charger**

The handheld unit will come with built in battery and a multi charger unit would be provided to charge multiple handsets simultaneously. 8 nos. of handheld units can be charged with each multi charger.

8. **Fixed units**

We propose to use TETRA radio with IP interface as fixed units. These units would connect directly to the TETRA network by way of dedicated LAN.

The unit would be essentially a touch screen panel PC running a typical TETRA despatcher software that would allow full communication and TETRA functionality. The unit would have a built in loudspeaker.

A dedicated headset with microphone would be provided along with a jack to plug it in.

For the LSO position a multiway audio jack with a selector switch would be provided so as to allow the operator to select either HDCS or Ship's Intercom or V/UHF for use with headset.

9. **Voice Logger**

A dedicated voice logger would be provided that would record all communications in the system. It would be possible to replay selective recordings at any time to review the communications.

Subject : Procurement of Helo Deck Communication System for VC 11190-91

Reference : Tender ref. no. MP/ 3400000297

BILL OF MATERIAL

Sr. No.	Item Description	Qty
1	TETRA Base Station	1 Nos.
2	TETRA fixed units	4 Nos.
3	IS TETRA mobile handsets with accessories	16 Nos.
4	Helmets	16 Nos.
5	Multi charger	2 Nos.
6	Whip aerial	1 Nos
7	Set of cables	1 Set
8	Recording & replay processor	1 Set
9	On Board Spares	1 Set

Final Part No. and description shall be provided prior to placement of order.

Yours faithfully,

For Elcome Integrated Systems Pvt. Ltd.,



Prithvijit Gupta
Sr. Manager (Sales & Projects)